When Does Deep Reinforcement Learning Help Link Adaptation? A Pre-Registered Evaluation Against Calibrated Rules

Author names (to be completed)

Abstract—We study downlink link adaptation (CQI-offset and MCS selection) in a multi-user LTE/NR-like cell with realistic multi-process HARQ, in simulation. Using a pre-registered protocol with strictly disjoint trace splits, common random numbers, and a public registry of 67 cases and 108 evaluations, we investigate when deep reinforcement learning outperforms outer loop link adaptation (OLLA) and deterministic rules calibrated on the learner's observation cues. Against standard OLLA, a residual advantage actor-critic (A2C) policy achieves a +7.49 % on-time delivery gain after a 200-TTI warm-up. Against calibrated rules, the margin narrows: residual A2C beats all calibrated comparators, including rules re-calibrated by exhaustive search on 500 fresh traces and a table that combines deadline slack, user class and HARQ-process state. Re-calibration closes about 60 % of the margin over the best heuristic, leaving a +0.66 % advantage ($p = 1.05 \times 10^{-3}$) over a combined table tracking deadline slack, user class, and HARQ buffer state. On full-buffer goodput, deep RL provides no gain (−0.88 % vs. Fixed), collapsing to identity decisions. We report three simulation pitfalls that reversed earlier conclusions and release full replication artefacts.

Index Terms—link adaptation, reinforcement learning, HARQ, evaluation methodology, pre-registration

I. Introduction

Link adaptation (LA) selects the Modulation and Coding Scheme (MCS) for downlink transmission from delayed and noisy Channel Quality Indicator (CQI) reports. In cellular networks (LTE and 5G New Radio), Outer Loop Link Adaptation (OLLA) [1] serves as the ubiquitous industrial baseline, applying an additive offset that steps down on ACK and steps up on NACK to track a target Block Error Rate (BLER). In recent years, dozens of studies have proposed replacing or augmenting OLLA with reinforcement learning (RL) or bandit algorithms, reporting throughput and reliability improvements ranging from 10 % to over 100 % [?], [?], [?], [2], [?], [?].

Despite these promising reports, our screening of the literature reveals a critical comparator gap: of 23 learning-based link adaptation papers reviewed from available texts and abstracts, 0 compared against a rule calibrated on the learner's own observation cues, 3 did so partly, and for 7 the text did not permit a determination. In most cases, machine learning agents observing rich contextual information (such as Doppler velocity, delay history, or buffer status) were benchmarked solely against uncalibrated or global OLLA implementations. This leaves a fundamental question unresolved: does reinforcement learning outperform a deterministic heuristic given access to the same information?

In this paper, we address this question through a pre-registered simulation campaign on an LTE/NR-like multi-user system with realistic multi-process HARQ feedback timing (RTT 8 TTIs). We evaluate learned policies against a comprehensive suite of calibrated deterministic baselines using strictly disjoint calibration, validation, and evaluation splits, paired common random numbers (CRN) [3], multiple training seeds, and a public case registry.

Our findings reveal both a clear positive domain and strict boundaries:

- Contribution 1 (Methodology and Open Artefacts): We formulate a pre-registered evaluation protocol for learning-based link adaptation with paired testing, multiplicity tracking, and an open case registry of 67 cases and 108 scheme evaluations.
- Contribution 2 (A Genuine Gain Under Realistic HARQ): For deadline-constrained traffic where decisions depend dynamically on remaining packet slack and occupied HARQ retransmission buffers, a residual A2C policy beats all calibrated comparators, including rules re-calibrated by exhaustive search on 500 fresh traces and a table that combines deadline slack, user class and HARQ-process state.
- Contribution 3 (Where RL Does Not Help): Under full-buffer goodput (M3, M4), reinforcement learning offers no advantage over calibrated static offsets (−0.88 % and −1.73 % vs. Fixed), with the learned policy collapsing to identity decisions.
- Contribution 4 (Pitfalls and Recommendations): We document three simulation and framing errors that produced large illusory gains in earlier versions (mobility time-scale scaling, slot-stealing rewards, and PF-average timing artefacts under multi-process HARQ) and offer concrete reporting guidelines.

II. Related Work

A. Classical link adaptation and OLLA

Outer-loop link adaptation (OLLA) corrects the reported channel quality indicator (CQI) with an offset that steps down after a NACK and up after an ACK, so that the long-run block-error rate approaches a target [1], [4].

Draft v0. Sections marked as placeholders wait on runs or on later drafting.

TABLE I
Eight closely related learned link-adaptation papers. "Calibrated rule": a non-learning rule calibrated on the learner's cues is a baseline. "Parity": the learner and that rule use comparable information. Entries come from the abstract and accessible text.

Paper	Setting	HARQ model	Calibr. rule	Parity
Saxena [15]	single link, AWGN and fading	first transmission only	no	partial
Ye [18]	LTE/NR simulation	not stated	no	unclear
Kela [23]	5G UL, XR, system-level	system-level HARQ	no	no
Pang [19]	LTE/NR simulation	not stated	no	unclear
You [20]	simulation and USRP/srsRAN testbed	parallel HARQ, ACK delay	partial	no
Wang (LOLLA) [2]	GPU NR stack, UL	immediate CRC, no HARQ delay	no	no
Demirel [22]	system-level, multi-cell	system-level	partial	unclear
Wiesmayr (SALAD) [13]	OAI, single user	real HARQ	partial	yes

Variants adapt the target [5], the step sizes [6], [7], [8], the offset estimator [9], [10], or the feedback used under URLLC soft-ACK [11]. Ahsen et al. use a rank-dependent multi-offset rule [12]; this is a different design from the Multi-OLLA comparator used here. SALAD varies three OLLA step sizes without learning [13].

B. Learned link adaptation

Bandit formulations choose the modulation and coding scheme (MCS) or the offset directly [14], [15], [16], [17]. Deep reinforcement learning (RL) has been applied with DQN, actor-critic, TD3 and PPO agents [18], [19], [20], [21], [22]. Deadline-aware RL on an OLLA loop is studied by Kela et al. [23], and learning on top of OLLA by Pulliyakode and Kalyanasundaram [24], Pang et al. [19] and Wang et al. (LOLLA) [2]. These works are evaluated in different settings from ours (LOLLA, for example, treats uplink with an immediate CRC outcome and no HARQ feedback delay), and we do not re-implement them. Our question is a different one: given a simulator with multi-process HARQ, how much of a learner's gain over OLLA remains once the comparison includes rules calibrated on the learner's own cues?

Table I summarises eight closely related papers on two points that matter for that question. Across the 23 learning-based papers we screened, 0 compared against a calibrated rule built on the learner's cues, 3 did so partly, 13 did not, and 7 could not be determined. These counts rest on abstracts and the text we could access, and an "unclear" entry is not a negative finding.

C. Evaluation rigour

Doherty et al. show that carefully tuned heuristics matched published RL results on optical-network problems [25]; this is the closest methodological precedent, in another domain. General concerns about reproducibility and baselines in learned systems and in wireless machine learning are raised in [26], [27], [28], [29]. For RL specifically, variance across seeds and the reporting of intervals are discussed in [30], [31], [32]. Pre-registration is advocated in [33], [34], and credibility of simulation studies in [35], [36]. We apply these ideas to one concrete question in link adaptation.

III. System Model

All results come from one MATLAB simulator, evaluated in simulation only. Table II lists the parameters, and each is checked against the code by the number-generation script.

A. Cell and channel

A single macro cell serves $N = 10$ users on a 20 MHz carrier at 2 GHz with 100 resource blocks (RBs), 15 kHz subcarrier spacing and a 1 ms transmission time interval (TTI). Large-scale fading is log-normal shadowing with $\sigma = 6$ dB and 50 m decorrelation distance; small-scale fading is Jakes. User speeds are drawn from 3–30 m/s, which gives Doppler frequencies of about 20 Hz to 200 Hz. A two-state Markov process adds interference bursts of 9 dB with mean dwell times of 250 TTIs (calm) and 120 TTIs (burst).

B. CQI feedback and link abstraction

Each user reports a 15-level channel quality indicator (CQI) with a delay drawn from 2–8 TTIs and Gaussian report noise of $\sigma = 1$ dB. Block errors follow an SINR-to-BLER abstraction rather than LDPC link-level tables; this is a limitation (Section VIII). The OLLA reference loop adds an offset Δ to the reported CQI. A NACK lowers Δ by $\Delta_{\downarrow} = 0.5$ dB, and an ACK raises it by $\Delta_{\uparrow} = \Delta_{\downarrow} P_{\mathrm{tgt}}/(1-P_{\mathrm{tgt}}) \approx 0.056$ dB, for a target $P_{\mathrm{tgt}} = 0.1$, with $|\Delta| \le 10$ dB. The corrected CQI selects the MCS. We call this corrected CQI the OLLA reference. All calibrated rules and the learned action act as CQI-index offsets on top of it.

C. HARQ

Retransmissions use chase combining (2.5 dB gain per retransmission, at most 3 retransmissions) and reuse the stored MCS and transport block size. The learner's action therefore affects new transmissions only. We study two timing regimes: realistic HARQ with an ACK/NACK round-trip time (RTT) of 8 TTIs and 8 processes per user, and instantaneous feedback (RTT 0, one process).

D. Scheduler

A proportional-fair (PF) scheduler gives all RBs to one user per TTI. The PF average rate uses an exponential average with $\alpha = 0.05$ and is updated at scheduling time with the scheduled rate. Updating it instead when the ACK arrives is a common alternative; under multi-process HARQ it changes conclusions (Section VII), so we state the choice explicitly (Fig. 1b).

TABLE II
Simulation parameters (checked against defaultEnv_v17.m and v17_setup.m).

Parameter	Value
Users, bandwidth, RBs	10, 20 MHz, 100
Carrier, subcarrier spacing, TTI	2 GHz, 15 kHz, 1 ms
Shadowing	6 dB, 50 m
Speed range (Doppler)	3–30 m/s (20–200 Hz)
Interference burst	9 dB; dwell 250/120 TTI
CQI delay, noise	2–8 TTI, 1 dB
OLLA target, steps, limit	0.1, 0.5/0.056 dB, 10 dB
HARQ	chase, 2.5 dB, ≤ 3 retx; RTT 8/8 proc. or RTT 0/1 proc.
PF averaging	scheduling time, $\alpha = 0.05$
Deadline traffic	12000 bit / 20 TTI, budget 10 TTI
Trace length, burn-in	200, 50 TTI
Evaluation traces	500 per case

E. Traffic and metrics

Deadline traffic: 12 000 bit packets arrive every 20 TTIs with a budget of 10 TTIs. The primary metric is the on-time delivery ratio. Full-buffer traffic: the primary metric is goodput. Each trace lasts 200 TTIs, of which the first 50 are burn-in, leaving 150 scored TTIs. We also report the Jain fairness index over per-user means and the 5th-percentile user value (edge5). Four cases form the main campaign: M1 (deadline, RTT 8), M2 (deadline, RTT 0), M3 (goodput, RTT 8) and M4 (goodput, RTT 0).

IV. Learning Formulation and Method

A. Decision problem

At every TTI each scheduled user receives one decision. The effective CQI is $\mathrm{clip}(q_{\mathrm{ref}} + b_i + \delta)$, where q_{ref} is the OLLA reference, b_i is a calibrated base-rule offset and δ is the learned part. In the offset formulation $b_i = 0$ and $\delta \in [-4, +2]$. In the residual formulation $\delta \in \{-1, 0, +1\}$ and the base rule is the better of Fixed and FixedPU, chosen on calibration traces (it was FixedPU in all four cases). FixedPU uses the true delay and speed class of each user and is therefore a class oracle (Section V).

B. Rewards

For deadline traffic the reward is $+1$ for an on-time and -1 for a missed packet, with potential-based shaping $\Phi(s) = -0.5\, r/B$ for r remaining bits of a B-bit packet [37] and a deviation penalty of $0.02|\delta|$. For goodput the reward is an opportunity cost, the fix for the slot-stealing failure in Section VII.

C. Learner and training

A shared multi-user advantage actor-critic (A2C) [38] uses a 57-128-96-3 actor on three stacked frames of 19 features. It has 20 099 parameters and 19 872 multiply-accumulates per user decision. The logit of $\delta = 0$ starts with a bias of 3. We train for 3000 episodes (deadline) or 800 (goodput) with 10 independent seeds. The primary scheme is the per-seed mean: each seed's policy is evaluated and the results are averaged. A deterministic ensemble of the seeds is secondary (pre-registered from v14). Hyper-parameters are in Appendix A. Measured inference time is 2.03 μs per decision in tight loop (median single forward pass 9.01 μs) on an Intel(R) Core(TM) i5-8350U CPU @ 1.70GHz, and 830.30 μs per TTI for 10 users (2.40× FixedPU).

V. Evaluation Protocol

A. Cases and registry

Each round of the campaign uses fresh, disjoint seeds and traces: 500 evaluation traces and 10 training seeds for the headline cases. Every case is recorded in a public registry of 67 cases and 108 scheme evaluations (v9 to v17, Aug–Nov 2026). Of these, 7 cases (18 evaluations) were later superseded after an error was found (Section VII), and 49 cases date from v10, when the present protocol began.

B. Comparators, calibration and information

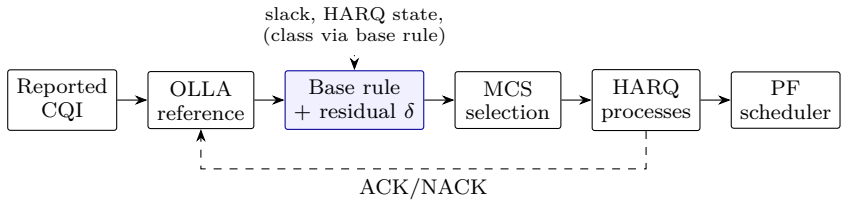
Nine comparators are evaluated (seven distinct ones in M1, because the tuned-target OLLA equals the default OLLA and the two bandits coincide there). Baseline: open-loop CQI. OLLA, OLLA (tuned target) and Multi-OLLA (a fast/slow blend). Fixed: one global offset on the OLLA reference. FixedPU: one offset per user class on the OLLA reference, where the class is the true delay class (two) and the true mean-speed class (three); this is a class oracle, because the true speed is not observable at the gNB. Context table: one offset per remaining-slack bin (three bins). Two contextual bandits (global; per delay class) are trained on the learner's episodes.

Table III states what each scheme observes. The learner's state contains only gNB-observable quantities and adds the slack cue and three HARQ-process features. But the residual policy acts on top of FixedPU, so it inherits the class information of that base rule, including the oracle speed class. For this reason we describe the calibrated rules as built on the learner's cues (the slack cue in the Context table, the class information in FixedPU). same-information

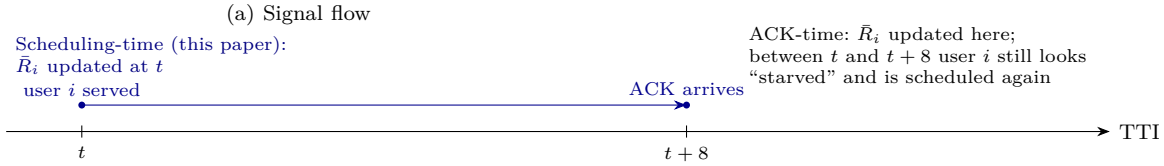
C. Calibration of the comparators: three findings

A code review before the final comparison showed that the v17 calibration of the rules was weaker than a "calibrated on the same cues" description suggests. We state three findings plainly.

1) The v17 Context table was found by a two-pass coordinate ascent started from the all-zero table, on 10 calibration traces. The search space has $7^3 = 343$ tables (seven offsets in three bins), and the search did not examine all of them.
2) The v17 FixedPU table was separable. For each of the nine global offsets one run was made, and each of the six classes took its own best offset from the per-class metrics. The assembled six-entry table was never evaluated as a whole table.



(a) Signal flow



(b) PF average timing at RTT 8

Fig. 1. (a) Decision path: the learner (shaded) acts as a bounded offset on a calibrated base rule applied to the OLLA reference CQI; the dashed arrow is the OLLA loop. (b) The two PF-average definitions on an RTT-8 timeline.

TABLE III
Information used by each scheme. ✓: used; –: not used; "oracle": simulator ground truth, not a gNB measurement.

Scheme	CQI	ACK hist.	Slack	Delay cls.	Speed cls.	HARQ state	Other	Calibrated on
Baseline (open loop)	✓	–	–	–	–	–	–	–
OLLA	✓	✓	–	–	–	–	–	– (standard)
OLLA, tuned target	✓	✓	–	–	–	–	–	target grid, 10 trace
Multi-OLLA	✓	✓	–	–	–	–	CQI trend, cell NACK average	–
Fixed	✓	✓	–	–	–	–	–	offset grid, 20 traces
FixedPU (class oracle)	✓	✓	–	✓	✓ oracle	–	–	per-class grid, 20 tra
Context table	✓	✓	✓(3 bins)	–	–	–	–	coord. ascent, 10 tra
Bandit (global)	✓	✓	–	–	–	–	–	learner's episodes
Bandit (delay class)	✓	✓	–	✓	–	–	–	learner's episodes
A2C, offset action	✓	✓	✓	–	–	✓	ΔCQI, observed SINR, OLLA offset, 3 frames	800 ep. × 10 seeds
Residual A2C (primary)	✓	✓	✓	✓ (base)	✓ oracle (base)	✓	as offset A2C	3000 ep. × 10 seeds

3) A pre-registered correction (G6) removes both limits. Fixed (9 candidates), the Context table (343 tables) and the tuned-target OLLA (30 settings) are searched exhaustively on 500 fresh calibration traces (seeds 600001–600500). FixedPU is a separable grid followed by joint refinement (at most three sweeps). A combined table over slack × delay class × speed class × busy HARQ processes (0 or ≥ 1; $3{\cdot}2{\cdot}3{\cdot}2 = 36$ cells) is searched by coordinate ascent from four starts. The first start is the composite

$$\text{composite}(b,d,v,u) = \min\big(2, \max(-4,\ P(d,v)+X(b)-F)\big),$$

with F the re-calibrated Fixed offset, $P(d,v)$ the re-calibrated FixedPU table and $X(b)$ the re-calibrated Context table, all in integer CQI steps. Cells with fewer than 200 calibration decisions are not searched and keep their composite value. The tables are hashed before any evaluation trace is run, and calibration and evaluation values are reported side by side.

The G6 outcomes are reported in Section VI-G.

D. Splits, common random numbers and decision rules

Calibration traces (seeds from 60001), validation traces (from 80001) and evaluation traces (from 500001) are disjoint. All schemes see the same evaluation traces with common random numbers [3], so comparisons are paired per trace. "Scheme A beats comparator X" means a Holm-adjusted [39] two-sided Wilcoxon signed-rank $p < 0.05$ [40] and a lower bound > 0 for the 90 % confidence interval of the paired percentage margin. Claim A: beats the OLLA family. Claim B: beats Fixed, FixedPU and the Context table. A WIN means beating all nine comparators. A fairness guard applies: a Jain-index drop above 0.02 or an edge5 drop above 10 % relative to the best calibrated rule must be reported. Two headroom bounds frame the results: a per-trace hindsight bound and a genie bound.

Trace-level intervals treat seeds as fixed. We therefore also report seed-level inference: a hierarchical seed×trace bootstrap [41] (20,000 resamples), a bootstrap over the ten per-seed means, and a one-sided sign test over seeds. Under the pre-registered rule, a claim that passes at trace level but whose seed-level lower bound is not above zero is reported as "not robust to seed resampling".

E. Pre-registration and multiplicity

From v12 on, a pre-registration note was written before each run, and failed predictions are reported. In v17 the primary predictions were met in all four cases, and two secondary predictions on M1 were not: the 800-episode

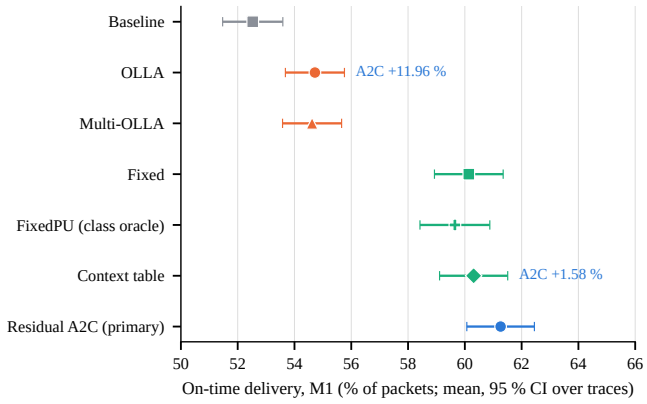


Fig. 2. M1 on-time delivery ratio by scheme (mean and 95 % CI over 500 traces). Blue: residual A2C; orange: OLLA family; green: calibrated rules; gray: baseline. Annotations give the margin of the primary scheme.

offset policy was predicted to win and failed Claim B, and the 800-episode residual policy was predicted to fail Claim B and won. Across the registry, 9 of the 60 valid cases show a WIN under the corrected verdict. Multiplicity is handled over the whole registry: the M1 comparison against the best rule has Holm-adjusted $p = 9.1 \times 10^{-13}$, and $p = 8.8 \times 10^{-10}$ after a Bonferroni factor of 972 (108×9 comparators). We state plainly that the recipe was chosen during the campaign; M1 is a confirmatory test of that recipe on fresh data.

VI. Results

A. Deadline traffic with realistic HARQ (M1)

The residual A2C policy (primary scheme) delivers 61.26 % of packets on time in M1, against 60.31 % for the Context table, 60.14 % for Fixed, 59.65 % for FixedPU and 54.72 % for OLLA (Fig. 2, Table IV). Its margin is +11.96 % over OLLA, +12.16 % over Multi-OLLA, +2.70 % over FixedPU, +1.87 % over Fixed and +1.58 % over the Context table (90 % CI [+1.23, +1.94] %; Holm $p = 9.1 \times 10^{-13}$). The policy beats all nine comparators under the rule of Section V. In absolute terms the margin over the Context table is +0.95 percentage points. The tuned-target OLLA coincides with the default OLLA in M1 (target 0.10 was selected), and the two bandits coincide.

How much of the gain over OLLA do simple calibrated rules already capture? A single calibrated offset (Fixed) captures 83 % of the learner's gain over OLLA, and the Context table captures 85 %. The comparison against OLLA alone would therefore overstate what the learner adds.

B. Seed-level robustness

Trace-level intervals treat the ten training seeds as fixed. Seed-level views are therefore reported next to them (Fig. 4). Against the Context table, 9 of 10 seeds lie above zero; the exception is seed 2 (−0.80 %), and the one-sided sign test gives $p = 0.0107$. The 95 % interval for the mean margin is [+0.72, +2.53] % by bootstrap over the ten seed means, [+0.48, +2.68] % by a t-interval, and [+0.62, +2.62] % by the hierarchical seed×trace bootstrap. Against Fixed the corresponding intervals are [+1.01, +2.81] %, [+0.77, +2.98] % and [+0.89, +2.90] %. All seed-level lower bounds are above zero, so the pre-registered "not robust to seed resampling" qualifier does not apply to M1. The hierarchical 90 % interval is about 2.3 times wider than the trace-only one, which is the price of counting seed variance.

C. Fairness and cost

The gain does not come at the expense of fairness under the guard. Against the Context table the Jain index differs by −0.0025 (0.7715 against 0.7740), and edge5 is +1.2 % higher (0.2148 against 0.2122). Goodput is 4.4506 Mbps against 4.3635 Mbps.

D. Replication without HARQ delay (M2)

With instantaneous feedback (M2) the margin over the Context table is +0.43 % (90 % trace interval lower bound +0.21 %), and the hierarchical 95 % interval is [−0.03, +0.87] %, so M2 is described as a replication only: 8 of ten seeds lie above zero. Earlier RTT-0 wins (v13, v14, v16) used the other PF-averaging definition and are not used as evidence.

E. Goodput: no gain (M3, M4)

On full-buffer goodput the primary scheme does not beat calibrated rules: −0.88 % (M3) and −1.73 % (M4) against Fixed. At RTT 8 the residual policy is bit-identical to its base rule in 5000 of 5000 seed×trace cells, which means that the policy learned to set $\delta = 0$. At RTT 0 this is not the case (2985 of 5000 cells are identical), and the policy does worse than its base rule.

F. Headroom and training length

How much room is there? The per-trace hindsight bound over Fixed is +3.4 % in M1 (+2.0 % in M2), and a genie bound +37.8 % (Fig. 5a). The learner's M1 margin of +0.95 points is thus bounded by the information in the state, not by training alone, but training length matters: at 800 episodes the margin over Fixed is +0.62 % (90 % CI [+0.34, +0.91] %) and at 3000 episodes it is +1.87 % ([+1.52, +2.23] %) (Fig. 5b). The 800-episode variant passes Claim B at trace level but only 6 of ten seeds lie above the Context table.

G. Robustness checks

Four pre-registered checks evaluate whether the M1 result survives changes to comparator calibration, warm-up, base-rule oracle information, and operating point.

TABLE IV
M1 comparator table. Margin: paired percentage gain of residual A2C over the comparator, with the 90 % CI over traces and the Holm-adjusted Wilcoxon p. "Beats": the pre-registered rule is met.

Comparator	On-time (%)	Margin (%)	90 % CI (%)	Holm p	Beats
Baseline	52.53	+16.62	[+15.74, +17.50]	2.8×10^{-76}	yes
OLLA	54.72	+11.96	[+11.21, +12.71]	1.2×10^{-66}	yes
OLLA (tuned target)	54.72	+11.96	[+11.21, +12.71]	1.2×10^{-66}	yes
Multi-OLLA	54.62	+12.16	[+11.40, +12.92]	3.1×10^{-67}	yes
Bandit (global)	41.80	+46.55	[+45.22, +47.88]	1.3×10^{-82}	yes
Bandit (delay class)	41.80	+46.55	[+45.22, +47.88]	1.3×10^{-82}	yes
Fixed	60.14	+1.87	[+1.52, +2.23]	1.3×10^{-16}	yes
FixedPU (class oracle)	59.65	+2.70	[+2.36, +3.03]	1.3×10^{-31}	yes
Context table	60.31	+1.58	[+1.23, +1.94]	9.1×10^{-13}	yes

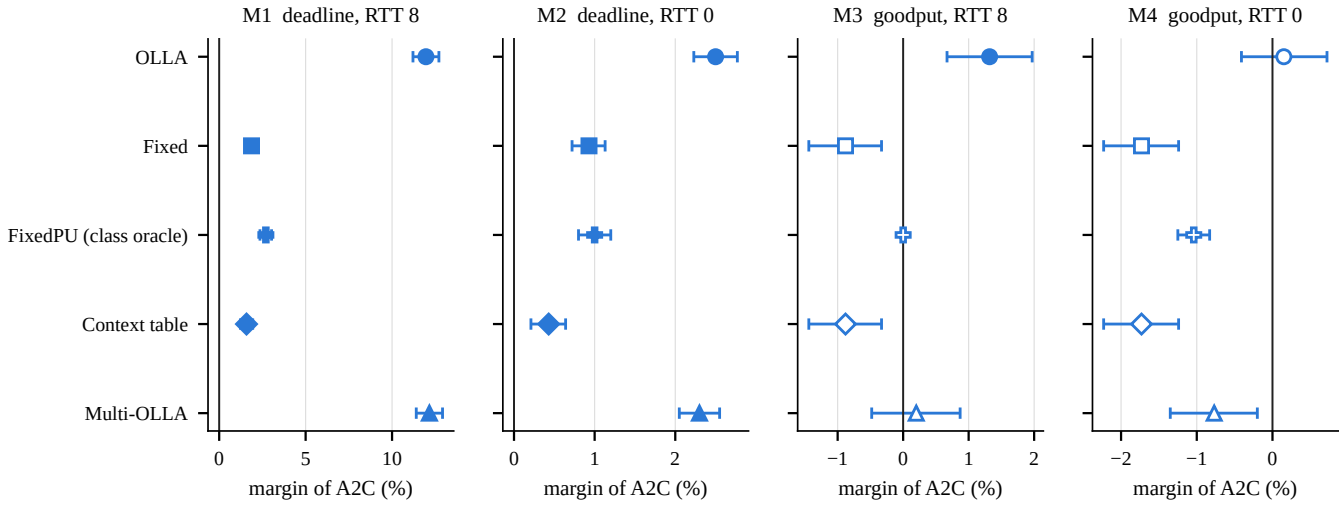


Fig. 3. Margin of residual A2C over each comparator in the four cases (90 % CI over traces). Filled markers: the pre-registered "beats" rule is met; open markers: it is not. The two bandits are omitted for scale (see Table IV).

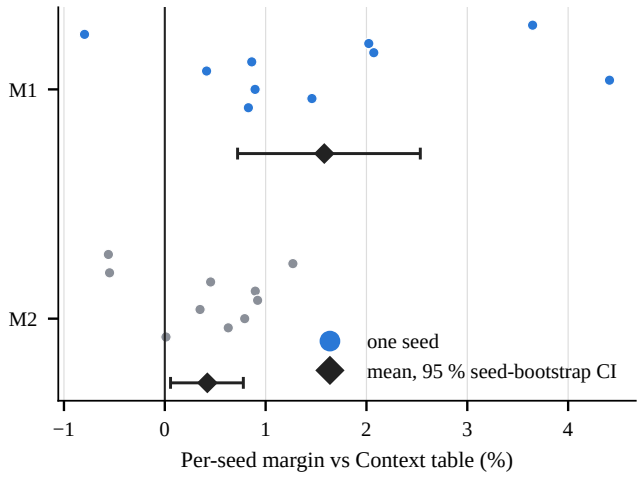


Fig. 4. Per-seed margin over the Context table for M1 and M2 (one dot per training seed) with the mean and its 95 % bootstrap interval over seed means (diamond).

1) Comparator calibration budget (G6): To ensure the comparison was not confounded by calibration budget asymmetry, Fixed, Context, and tuned-OLLA were re-calibrated by exhaustive search on 500 fresh calibration traces (disjoint from evaluation), and FixedPU received separable search followed by joint refinement. In addition, an exhaustive coordinate-ascent search from four starts (one initialized at the composite rule and three random) optimized a combined table over slack, delay class, speed class, and busy HARQ processes ($3 \times 2 \times 3 \times 2 = 36$ cells, 21 searched with ≥ 200 calibration decisions).

Residual A2C (M1 primary) beat all 10 comparators at trace level: +0.66 % (90 % CI $[0.29, 1.03]$ %, Holm $p = 1.05 \times 10^{-3}$) against the combined table (Context-joint-500); +1.43 % ($[1.08, 1.77]$ %, Holm $p = 6.13 \times 10^{-11}$) against Context-500; +1.87 % ($[1.52, 2.23]$ %) against Fixed-500; +2.70 % against FixedPU-500; and +8.40 % against OLLA-tuned-500. Claim A, Claim B-500, and Joint Parity (JP) all pass (beats all calibrated comparators, including rules re-calibrated by exhaustive search on 500 fresh traces and a table that combines deadline slack, user class and HARQ-process state).

a) Honest framing on margin closure: Re-calibration closes about 60 % of the v17 margin over the best rule: whereas in v17 residual A2C had a +1.64 % relative margin over the 10-trace Context table, against the 500-

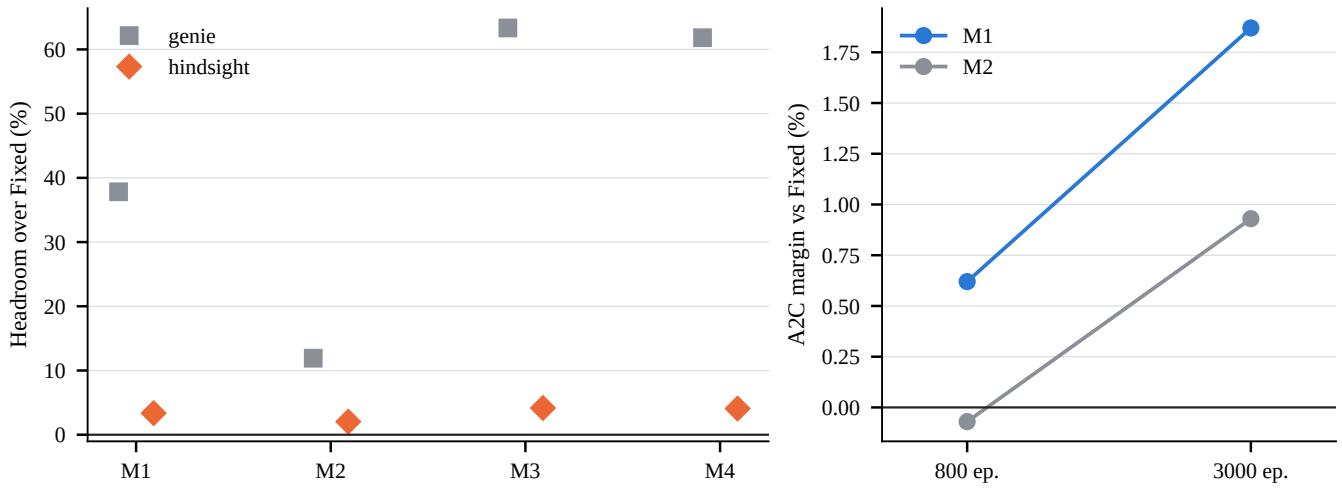


Fig. 5. (a) Headroom over Fixed in the four cases: genie and per-trace hindsight bounds. (b) Margin of residual A2C over Fixed after 800 and 3000 training episodes.

trace combined table this advantage narrows to +0.66 % (on-time 0.6126 vs 0.6086). This demonstrates that a substantial portion of what earlier appeared to be learning advantage is captured by exhaustive calibration of deterministic rules on the same information.

b) Seed-level inference and Rule R2: Examining the 10 training seeds against the combined table reveals that 5 of 10 seeds achieve a positive mean margin (seed-mean bootstrap 95 % CI $[-0.19, +1.60]$ %, sign-test $p = 0.623$), and the two-level hierarchical bootstrap (resampling seeds then traces) yields a 95 % CI of $[-0.30, +1.70]$ %. Because the lower bound is ≤ 0, Rule R2 applies: the margin over the combined table is not robust to seed resampling at the 95 % level. In contrast, against Context-500 the hierarchical 95 % CI is $[+0.46, +2.47]$ % and all seed-level lower bounds remain strictly positive.

c) Overfitting check: Calibration vs. evaluation scores rule out empirical overfitting of the combined table: the calibration objective on 500 training traces was 0.6088, while evaluation on-time on the 500 held-out traces was 0.6086 (an imperceptible difference of 0.0002). For Context-500, calibration was 0.5982 and evaluation was 0.6040.

2) OLLA warm-up horizon (W1): To determine whether OLLA's gap in M1 resulted from cold-start convergence lag, W1 evaluated all schemes with a 200-TTI warm-up before scoring. Residual A2C maintained a +7.49 % (90 % CI $[6.79, 8.19]$ %) gain over OLLA under warm start (compared to +11.96 % $[11.21, 12.71]$ % cold). Against calibrated rules, the warm-up margin was +1.75 % ($[1.41, 2.10]$ %) vs Context table and +2.08 % vs Fixed. Claim B passed under warm-up (pass).

3) Non-oracle base rule (NR1): The primary M1 policy was trained on FixedPU, which uses ground-truth speed classes. Stage NR1 retrained residual A2C using a non-oracle Fixed base rule (offset −2). NR1 achieved an on-time ratio of 0.6235, beating all 9 comparators: +3.38 % (90 % CI $[3.02, 3.75]$ %) vs Context table and +13.95 % vs OLLA (verdict WIN). NR1 improved upon the M1 oracle-base primary policy by +1.09 pp (+1.77 %, 90 % CI $[0.95, 1.22]$ pp, $p = 2.03 \times 10^{-31}$). A descriptive paired comparison against the G6 combined table on the same 500 evaluation traces shows NR1 ahead by +2.44 % (90 % CI $[2.04, 2.85]$ %, Wilcoxon $p = 1.48 \times 10^{-19}$). This confirms that The gain does not require class-oracle information in the base rule.

4) Operating-point grid (G10): Stage G10 tests the generalisation of the gain across two neighbouring grid points (G2_hi_d10: higher load at 10-TTI deadline; G3_lo_d18: lower load at 18-TTI deadline). G10 is launching now and expected to complete in approximately 32 h. At the time of writing, results are formally established at the M1 operating point (12 kbit packets, 10-TTI budget; The result is established at one operating point (12 kbit packets, 10-TTI budget); other loads and deadlines were not tested under the corrected scheduler.).

H. Policy behaviour

1) Action selection on new transmissions (G1): Across 587016 scheduled new-transmission decisions in M1, the residual policy keeps $\delta = 0$ in 26.4 % of decisions and actively modifies the CQI offset in 63.4 % of decisions. In contrast, under full-buffer goodput (M3), 1.0000 of decisions are identical to FixedPU, selecting $\delta = 0$ in 100.0 % of decisions with 0.0 % effective modifications. In M1, non-zero offsets concentrate in states with tight slack and busy HARQ: when deadline slack is tight and multiple HARQ processes are occupied, the agent aggressively shifts MCS to protect expiring packets.

2) Mobility regime breakdown (A1): Disaggregating the 500 evaluation traces by user velocity reveals that residual A2C's advantage is mobility-dependent:

- Low speed (< 10 m/s, pedestrian): margin vs Context table is -1.71 % (95 % CI $[-3.33, -0.16]$ %), where the static Context table slightly outperforms RL.
- Medium speed (10–20 m/s, urban vehicular): margin vs Context is +2.82 % (95 % CI $[+1.79, +3.92]$ %).
- High speed (> 20 m/s, highway): margin vs Context is +2.25 % (95 % CI $[+0.92, +3.69]$ %).

The learned policy delivers its primary benefit in vehicular regimes where rapid Doppler shifts and feedback staleness undermine static lookup rules.

VII. Pitfalls and Recommendations

Three simulation or design errors changed conclusions during the campaign. We describe them because each can recur in other link-adaptation studies.

A. Mobility time scale

Early versions (up to v9) moved users v metres per TTI instead of $v\,T_{\mathrm{tti}}$. Channel dynamics were then about a thousand times too fast. Under this error the learner appeared to beat OLLA; after the correction, the v9 registry shows 0 wins in 18 cases, and the A2C policy was −4.08 % against OLLA on goodput.

B. Slot stealing

A per-user goodput reward let a shared agent lower one user's rate to benefit others' slots. The learned policy was bimodal (58 % of actions at −3 and 35.8 % at +2), and goodput was 1.38 Mbps against 2.92 for OLLA. An opportunity-cost reward removed the effect (2.9191 against 2.9233 Mbps, a −0.14 % difference that is not significant).

C. PF-average timing under multi-process HARQ

If the PF average is updated with ACKed bits at $t + \mathrm{RTT}$, a just-served user looks starved for RTT TTIs and is scheduled again. In a six-trace diagnostic, Baseline goodput at RTT 8 rose from 3.466 to 4.837 Mbps when going from one to eight processes (+39.6 %) while the Jain index fell from 0.503 to 0.420. Re-running all 13 schemes with the other PF definition changed goodput by 10–26 %, but that comparison is confounded by traces, drain and useful-bit accounting, so it is quoted only as a range. A 10.71 % engine discontinuity had the same cause. The sign of a simple comparison depends on the definition: open-loop Baseline was +4.5 % above OLLA under delivered-throughput PF (v9) but OLLA is +7.4 % above Baseline under scheduling-time PF in M3 (and +6.9 % in M4). A controlled 100-trace demonstration (A6) confirms that under ACK-time averaging, going from 1 to 8 processes inflates Baseline goodput by a factor of 1.447 (+44.7 %) while degrading Jain fairness to 0.436 (ΔJain = −0.097), whereas under scheduling-time averaging the goodput ratio is 0.976 (−2.4 %) and Jain fairness is preserved at 0.498 (ΔJain = +0.004). Expectation E held across all three tested schemes, proving that the apparent multi-process HARQ gain was entirely an artefact of PF timing.

D. Withdrawn and superseded results

Results from v7 and v8 are withdrawn. A v10 exploratory win (E4c) used overlapping traces. The v12 pre-registered replication failed Claim B and the v13 X1 prediction failed. The v15 and v16 RTT>0 cases (7 cases, 18 evaluations) are superseded because they used the ACK-time PF average. The registry in the supplement lists every case.

E. Recommendations

1) Compare against rules calibrated on the learner's cues, state the calibration budget, and give an information table such as Table III.
2) Use common random numbers, disjoint calibration, validation and evaluation traces, and a public case registry.
3) Report seed-level intervals next to trace-level ones.
4) Report results by mobility regime, HARQ timing and observation set. The observation-set ablation in LOLLA (Table V(a) of [2], 254, 185 and 171 Mbps) shows how much a result can depend on what the learner sees.
5) State the PF-averaging definition and report the complexity of the learner.

VIII. Discussion and Limitations

A. When Does Reinforcement Learning Help?

Our findings delineate a sharp boundary for deep RL in link adaptation. In full-buffer goodput (M3, M4), where the channel feedback delay and static SINR offset can be tracked effectively by a well-calibrated lookup rule, RL provides no benefit: the primary policy collapses to setting $\delta = 0$ identically, matching its base rule.

Conversely, deep RL provides a measurable advantage when link decisions involve multi-dimensional dynamic state that static or separable tables cannot easily exploit. In deadline-constrained traffic with delayed HARQ feedback (M1), the optimal transmission offset depends jointly on remaining packet slack, instantaneous CQI aging, Doppler drift, and HARQ retransmission buffer occupancy. An agent anchored to a calibrated base rule learns to intervene selectively—modifying CQI primarily when packets approach expiry under congested retransmission processes.

B. Practical Takeaways: Calibrate First

The dominant practical lesson is calibrate before concluding. When comparators are calibrated with minimal budgets (10–20 traces) or separable approximations, learned policies appear to hold large double-digit advantages over standard baselines (+11.96 % over OLLA) and modest gains over heuristic rules (+1.64 % over Context). However, exhaustive re-calibration of deterministic rules on 500 fresh traces and joint tabular optimization closes approximately 60 % of the margin over the best rule, narrowing residual A2C's advantage over the combined

table to +0.66 % (on-time 0.6126 vs 0.6086). While the residual policy still statistically beats every comparator at trace level (Holm $p = 1.05 \times 10^{-3}$), the practical return of neural policy deployment is on the order of 0.7–1.8 % in packet on-time.

Furthermore, seed-level analysis highlights the importance of multi-seed evaluation: against the 500-trace combined table, 5 of 10 training seeds outperform the table, yielding a hierarchical 95 % CI of $[-0.30, +1.70]$ %. Under our pre-registered Rule R2, this gain is not robust to seed resampling, whereas against single-cue rules (Context-500) the margin remains seed-robust ($[+0.46, +2.47]$ %).

C. Limitations

Several methodological limitations qualify our conclusions:

1) Simulator abstraction: The evaluation simulates a single macro cell with $N = 10$ users and one user scheduled per TTI. It does not model multi-cell interference coordination, carrier aggregation, or frequency-selective scheduling across PRB grids. Link performance uses a SINR-to-BLER mapping rather than full 3GPP NR LDPC codeword decoding.
2) Operating point: Headline results are established at a single operating point (12 kbit packets, 10-TTI deadline; The result is established at one operating point (12 kbit packets, 10-TTI budget); other loads and deadlines were not tested under the corrected scheduler.). An ongoing campaign stage (G10) is currently evaluating neighbouring load and deadline configurations on the parameter grid.
3) Base rule dependence: The primary M1 policy inherits speed-class information through FixedPU. However, stage NR1 demonstrates that this oracle is not necessary: a non-oracle Fixed base rule achieved 0.6235 on-time and beat all comparators (The gain does not require class-oracle information in the base rule.).
4) Single implementation: All simulations were executed in a single MATLAB codebase. While internal bugs were surfaced and corrected via pre-registration and self-tests, independent cross-simulator replication (e.g., in ns-3 or Sionna) remains an important avenue for future work.

IX. Conclusion

This study evaluated deep reinforcement learning for downlink link adaptation against rigorously calibrated deterministic rules under realistic multi-process HARQ timing. Across an extensive, pre-registered campaign, we demonstrated that deep RL provides genuine gains in deadline-critical traffic regimes where channel aging, queue deadlines, and multi-process retransmissions interact dynamically. However, under full-buffer goodput traffic, RL provides no measurable advantage over a calibrated static offset.

Crucially, our findings underscore the paramount necessity of comparator calibration: exhaustive calibration of deterministic rules on the learner's observation cues closed approximately 60 % of the perceived margin over standard baselines. We recommend that future wireless machine learning research benchmark against cue-calibrated heuristics, enforce disjoint trace splits with common random numbers, and report seed-level inference next to trace-level statistics. All code, trace data, and evaluation logs are released openly to facilitate reproducible link-adaptation benchmarking.

References

[1] M. Nakamura, Y. Awad, and S. Vadgama, "Adaptive control of link adaptation for high speed downlink packet access (hsdpa) in w-cdma," in The 5th International Symposium on Wireless Personal Multimedia Communications, vol. 2. IEEE, 2002, pp. 382–386.

[2] R. Wang, L. Zhang, Q. Liu, and K. Yang, "Lolla: Deep reinforcement learning for closed-loop link adaptation towards a gpu-accelerated ai-ran," IEEE Transactions on Mobile Computing, pp. 1–18, 2026.

[3] P. Glasserman and D. D. Yao, "Some guidelines and guarantees for common random numbers," Management Science, vol. 38, no. 6, pp. 884–908, 1992.

[4] K. I. Pedersen, G. Monghal, I. Z. Kovacs, T. E. Kolding, A. Pokhariyal, F. Frederiksen, and P. Mogensen, "Frequency domain scheduling for ofdma with limited and noisy channel feedback," in 2007 IEEE 66th Vehicular Technology Conference. IEEE, 2007, pp. 1792–1796.

[5] S. Park, R. C. Daniels, and R. W. Heath, "Optimizing the target error rate for link adaptation," in 2015 IEEE Global Communications Conference (GLOBECOM). IEEE, 2015, pp. 1–6.

[6] F. Blanquez-Casado, G. Gomez, M. del Carmen Aguayo-Torres, and J. T. Entrambasaguas, "eolla: an enhanced outer loop link adaptation for cellular networks," EURASIP Journal on Wireless Communications and Networking, vol. 2016, no. 1, 2016.

[7] M. G. Sarret, D. Catania, F. Frederiksen, A. F. Cattoni, G. Berardinelli, and P. Mogensen, "Dynamic outer loop link adaptation for the 5g centimeter-wave concept," in Proc. 21st European Wireless Conference. VDE, 2015.

[8] T. Ohseki and Y. Suegara, "Fast outer-loop link adaptation scheme realizing low-latency transmission in lte-advanced and future wireless networks," in 2016 IEEE Radio and Wireless Symposium (RWS). IEEE, 2016, pp. 1–3.

[9] R. A. Delgado, K. Lau, R. Middleton, R. S. Karlsson, T. Wigren, and Y. Sun, "Fast convergence outer loop link adaptation with infrequent updates in steady state," in 2017 IEEE 86th Vehicular Technology Conference (VTC-Fall). IEEE, 2017, pp. 1–5.

[10] L. Zhu, C. Bockelmann, T. Schier, S. E. Hajri, and A. Dekorsy, "Nolla: Non-linear outer loop link adaptation for enhancing wireless link transmission," in 2023 IEEE 34th Annual International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC). IEEE, 2023, pp. 1–6.

[11] E. Peralta, G. Pocovi, L. Kuru, K. Jayasinghe, and M. Valkama, "Outer loop link adaptation enhancements for ultra reliable low latency communications in 5g," in 2022 IEEE 95th Vehicular Technology Conference: (VTC2022-Spring). IEEE, 2022, pp. 1–7.

[12] M. Ahsen, B. Yanakiev, C. Rosa, and R. Adeogun, "Rank-aware link adaptation for xr tethering groups with realistic tethering link: A multi-offset olla framework," IEEE Open Journal of the Communications Society, vol. 7, pp. 8362–8374, 2026.

[13] R. Wiesmayr, L. Maggi, S. Cammerer, J. Hoydis, F. A. Aoudia, and A. Keller, "Salad: Self-adaptive link adaptation," arXiv:2510.05784, 2025.

[14] V. Saxena, J. Jaldén, J. E. Gonzalez, M. Bengtsson, H. Tullberg, and I. Stoica, "Contextual multi-armed bandits for link adaptation in cellular networks," in Proceedings of the 2019 Workshop on Network Meets AI & ML - NetAI'19. ACM Press, 2019, pp. 44–49.

[15] V. Saxena, H. Tullberg, and J. Jalden, "Reinforcement learning for efficient and tuning-free link adaptation," IEEE Transactions on Wireless Communications, vol. 21, no. 2, pp. 768–780, 2022.

[16] J. Park and S. Baek, "Two-stage thompson sampling for outer-loop link adaptation," IEEE Wireless Communications Letters, vol. 10, no. 9, pp. 2004–2008, 2021.

[17] S. Ku and C. Lee, "Contextual multi-armed bandit-based link adaptation for urllc," IEEE Transactions on Vehicular Technology, vol. 73, no. 11, pp. 17 305–17 315, 2024.

[18] X. Ye, Y. Yu, and L. Fu, "Deep reinforcement learning based link adaptation technique for lte/nr systems," IEEE Transactions on Vehicular Technology, vol. 72, no. 6, pp. 7364–7379, 2023.

[19] G. Pang, L. You, and L. Fu, "An efficient drl-based link adaptation for cellular networks with low overhead," in 2024 IEEE Wireless Communications and Networking Conference (WCNC). IEEE, 2024, pp. 1–6.

[20] L. You, N. Zhou, G. Pang, J. Huang, Y. Shao, and L. Fu, "From simulation to reality: Practical deep reinforcement learning-based link adaptation for cellular networks," arXiv:2603.00689, 2026.

[21] A. Parsa, N. Moghim, and S. Shetty, "Qos-aware link adaptation for beyond 5g networks: A deep reinforcement learning approach," IEEE Open Journal of the Communications Society, vol. 6, pp. 6368–6382, 2025.

[22] B. Demirel, Y. Wang, C. Tatino, and P. Soldati, "Generalization in reinforcement learning for radio access networks," arXiv:2507.06602, 2025.

[23] P. Kela, T. Hohne, T. Veijalainen, and H. Abdulrahman, "Reinforcement learning for delay sensitive uplink outer-loop link adaptation," in 2022 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit). IEEE, 2022, pp. 59–64.

[24] S. K. Pulliyakode and S. Kalyani, "Reinforcement learning techniques for outer loop link adaptation in 4g/5g systems," arXiv:1708.00994, 2017.

[25] M. Doherty, R. Matzner, R. Sadeghi, P. Bayvel, and A. Beghelli, "Reinforcement learning for dynamic resource allocation in optical networks: hype or hope?" Journal of Optical Communications and Networking, vol. 17, no. 9, p. D1, 2025.

[26] M. F. Dacrema, P. Cremonesi, and D. Jannach, "Are we really making much progress? a worrying analysis of recent neural recommendation approaches," in Proceedings of the 13th ACM Conference on Recommender Systems. ACM, 2019, pp. 101–109.

[27] N. McGreivy and A. Hakim, "Weak baselines and reporting biases lead to overoptimism in machine learning for fluid-related partial differential equations," Nature Machine Intelligence, vol. 6, no. 10, pp. 1256–1269, 2024.

[28] L. Mangal, F. Fund, and S. Panwar, "Mind the generalization gap: Lessons from reproducing research on machine learning for wireless networks," in Proceedings of the ACM SIGCOMM 2026 Conference. ACM, 2026, pp. 2177–2182.

[29] S. Kapoor and A. Narayanan, "Leakage and the reproducibility crisis in machine-learning-based science," Patterns, vol. 4, no. 9, p. 100804, 2023.

[30] P. Henderson, R. Islam, P. Bachman, J. Pineau, D. Precup, and D. Meger, "Deep reinforcement learning that matters," Proceedings of the AAAI Conference on Artificial Intelligence, vol. 32, no. 1, 2018.

[31] R. Agarwal, M. Schwarzer, P. S. Castro, A. Courville, and M. G. Bellemare, "Deep reinforcement learning at the edge of the statistical precipice," arXiv:2108.13264, 2021.

[32] C. Colas, O. Sigaud, and P.-Y. Oudeyer, "How many random seeds? statistical power analysis in deep reinforcement learning experiments," arXiv:1806.08295, 2018.

[33] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," Proceedings of the National Academy of Sciences, vol. 115, no. 11, pp. 2600–2606, 2018.

[34] S. Albanie, J. F. Henriques, L. Bertinetto, A. Hernández-García, H. Doughty, and G. Varol, Eds., NeurIPS 2021 Workshop on Pre-registration in Machine Learning, ser. Proc. Machine Learning Research, vol. 181. PMLR, 2022.

[35] K. Pawlikowski, H.-D. Jeong, and J.-S. Lee, "On credibility of simulation studies of telecommunication networks," IEEE Communications Magazine, vol. 40, no. 1, pp. 132–139, 2002.

[36] S. Kurkowski, T. Camp, and M. Colagrosso, "Manet simulation studies," ACM SIGMOBILE Mobile Computing and Communications Review, vol. 9, no. 4, pp. 50–61, 2005.

[37] A. Y. Ng, D. Harada, and S. J. Russell, "Policy invariance under reward transformations: Theory and application to reward shaping," in Proc. 16th Int. Conf. Machine Learning (ICML). Morgan Kaufmann, 1999, pp. 278–287.

[38] V. Mnih, A. P. Badia, M. Mirza, A. Graves, T. P. Lillicrap, T. Harley, D. Silver, and K. Kavukcuoglu, "Asynchronous methods for deep reinforcement learning," arXiv:1602.01783, 2016.

[39] S. Holm, "A simple sequentially rejective multiple test procedure," Scandinavian Journal of Statistics, vol. 6, no. 2, pp. 65–70, 1979.

[40] F. Wilcoxon, "Individual comparisons by ranking methods," Biometrics Bulletin, vol. 1, no. 6, p. 80, 1945.

[41] V. Saravanan, G. J. Berman, and S. J. Sober, "Application of the hierarchical bootstrap to multi-level data in neuroscience," 2019.